

Pregame talks

Math AA HL

Quiz 1 · Week 8: 2.2, 2.1, 1.4, 2.11, 2.12

Five questions, ten minutes

Time limit: 10 minutes, strict. Start a timer. Stop when it rings, finished or not. No answers during the quiz; they are on page 2 for afterwards.

What to send, by Saturday 10 October, before the Zoom: one email to Gauti.Eggertsson@gmail.com, subject “Quiz 1”, three lines:

Done at [time]. Got to problem [n]. Looked at answers for [which, or none].

Then the Zoom: the week’s sprints again, without the answer pages.

1. The vertex of $y = x^2 - 6x + 1$.
2. For which values of k does $x^2 + kx + 9 = 0$ have a repeated root?
3. Gradient and midpoint of the segment from $(-1, 2)$ to $(3, 10)$.
4. The remainder when $x^3 - 4x + 1$ is divided by $(x - 2)$.
5. The sum and the product of the roots of $3x^2 - 5x + 7 = 0$ (no need to find them).

Look only when stuck, and say which ones in the email. On the weekend these are redone without this page.

1. $x = -\frac{-6}{2} = 3$; $y = 9 - 18 + 1 = -8$. **Vertex** $(3, -8)$.

2. $\Delta = k^2 - 36 = 0$: $k = 6$ or $k = -6$.

3. $m = \frac{10 - 2}{3 - (-1)} = \frac{8}{4} = 2$; midpoint $(1, 6)$.

4. $f(2) = 8 - 8 + 1 = 1$.

5. Sum $= -\frac{-5}{3} = \frac{5}{3}$; product $= \frac{7}{3}$. ($\Delta = 25 - 84 < 0$: the roots are a conjugate pair, and Vieta does not care.)